

# GPIO & RTC GPIO

[\[中文\]](#)

## GPIO Summary

The ESP32-P4 chip features 55 physical GPIO pins (GPIO0 ~ GPIO54). Each pin can be used as a general-purpose I/O, or to be connected to an internal peripheral signal. Through GPIO matrix and IO MUX, peripheral input signals can be from any IO pins, and peripheral output signals can be routed to any IO pins. Together these modules provide highly configurable I/O. For more details, see *ESP32-P4 Technical Reference Manual > IO MUX and GPIO Matrix (GPIO, IO\_MUX)* [\[PDF\]](#).

The table below provides more information on pin usage, and please note the comments in the table for GPIOs with restrictions.

| GPIO   | Analog Function | LP GPIO   | Comments |
|--------|-----------------|-----------|----------|
| GPIO0  |                 | LP_GPIO0  |          |
| GPIO1  |                 | LP_GPIO1  |          |
| GPIO2  | TOUCH0          | LP_GPIO2  |          |
| GPIO3  | TOUCH1          | LP_GPIO3  |          |
| GPIO4  | TOUCH2          | LP_GPIO4  |          |
| GPIO5  | TOUCH3          | LP_GPIO5  |          |
| GPIO6  | TOUCH4          | LP_GPIO6  |          |
| GPIO7  | TOUCH5          | LP_GPIO7  |          |
| GPIO8  | TOUCH6          | LP_GPIO8  |          |
| GPIO9  | TOUCH7          | LP_GPIO9  |          |
| GPIO10 | TOUCH8          | LP_GPIO10 |          |
| GPIO11 | TOUCH9          | LP_GPIO11 |          |
| GPIO12 | TOUCH10         | LP_GPIO12 |          |
| GPIO13 | TOUCH11         | LP_GPIO13 |          |

| GPIO   | Analog Function | LP GPIO   | Comments      |
|--------|-----------------|-----------|---------------|
| GPIO14 | TOUCH12         | LP_GPIO14 |               |
| GPIO15 | TOUCH13         | LP_GPIO15 |               |
| GPIO16 | ADC1_CH0        |           |               |
| GPIO17 | ADC1_CH1        |           |               |
| GPIO18 | ADC1_CH2        |           |               |
| GPIO19 | ADC1_CH3        |           |               |
| GPIO20 | ADC1_CH4        |           |               |
| GPIO21 | ADC1_CH5        |           |               |
| GPIO22 | ADC1_CH6        |           |               |
| GPIO23 | ADC1_CH7        |           |               |
| GPIO24 |                 |           |               |
| GPIO25 |                 |           |               |
| GPIO26 |                 |           |               |
| GPIO27 |                 |           |               |
| GPIO28 |                 |           |               |
| GPIO29 |                 |           |               |
| GPIO30 |                 |           |               |
| GPIO31 |                 |           |               |
| GPIO32 |                 |           |               |
| GPIO33 |                 |           |               |
| GPIO34 |                 |           | Strapping pin |
| GPIO35 |                 |           | Strapping pin |
| GPIO36 |                 |           | Strapping pin |
| GPIO37 |                 |           | Strapping pin |
| GPIO38 |                 |           | Strapping pin |
| GPIO39 |                 |           |               |
| GPIO40 |                 |           |               |
| GPIO41 |                 |           |               |
| GPIO42 |                 |           |               |
| GPIO43 |                 |           |               |

| GPIO   | Analog Function                              | LP GPIO | Comments |
|--------|----------------------------------------------|---------|----------|
| GPIO44 |                                              |         |          |
| GPIO45 |                                              |         |          |
| GPIO46 |                                              |         |          |
| GPIO47 |                                              |         |          |
| GPIO48 |                                              |         |          |
| GPIO49 | ADC2_CH0                                     |         |          |
| GPIO50 | ADC2_CH1                                     |         |          |
| GPIO51 | ADC2_CH2, ANA_CMPR_CH0 reference voltage     |         |          |
| GPIO52 | ADC2_CH3, ANA_CMPR_CH0 input (non-inverting) |         |          |
| GPIO53 | ADC2_CH4, ANA_CMPR_CH1 reference voltage     |         |          |
| GPIO54 | ADC2_CH5, ANA_CMPR_CH1 input (non-inverting) |         |          |

### **Note**

- **Strapping pin:** GPIO34, GPIO35, GPIO36, GPIO37, and GPIO38 are strapping pins. For more information, please refer to [datasheet](#).
  - USB-JTAG: GPIO24 and GPIO25 are used by USB-JTAG by default. If they are reconfigured to operate as normal GPIOs, USB-JTAG functionality will be disabled.

There is also separate "RTC GPIO" support, which functions when GPIOs are routed to the "RTC" low-power, analog subsystem, and Low-Power(LP) peripherals. These pin functions can be used when:

- In Deep-sleep mode
- The [Ultra Low Power LP-Core co-processor](#) is running
- Analog functions such as ADC/DAC/etc are in use
- LP peripherals, such as LP\_UART, LP\_I2C, are in use

## **IO Configuration**

An IO can be used in two ways:

- As a simple GPIO input to read the level on the pin, or as a simple GPIO output to output the desired level on the pin.
- As a peripheral signal input/output.